STAT 559 – Note 04
Product Measures and Fubini–Tonelli

Edited transcription of handwritten lecture notes

These notes transcribe and edit the indicated handwritten PDF. Source-page markers preserve traceability. Yellow boxes retain original TODOs; red boxes explain mathematical corrections; blue boxes add explanations or proof details. The reference is *stats310notes.pdf* (provided in `notes/`); references below use its printed section numbers.

Source page(s): 1–4

1 Product measurable spaces

The motivation is to turn an integral on a product space into iterated one-dimensional integrals. For measurable spaces $(\Omega_i, \mathcal{F}_i)$, $i = 1, \dots, n$, let

$$\Omega = \prod_{i=1}^{n} \Omega_i, \qquad \mathcal{F} = \bigotimes_{i=1}^{n} \mathcal{F}_i = \sigma\left(\left\{ \prod_{i=1}^{n} A_i : A_i \in \mathcal{F}_i \right\} \right). \tag{1}$$

The generating sets are measurable *rectangles*. When $\Omega_i = \mathbb{R}$ and $\mathcal{F}_i = \mathcal{B}(\mathbb{R})$, this product space is $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$.

Original TODO: Why is the product Borel sigma-algebra the usual one?

For $n = 2$, each Borel rectangle belongs to $\mathcal{B}(\mathbb{R}^2)$: its coordinate factors are inverse images under continuous projections. Conversely every open subset of $\mathbb{R}^2$ is a countable union of open rectangles with rational endpoints, so it belongs to $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$. The two inclusions prove equality. The same countable-base argument works in finite dimensions (reference Exercise 3.1.3).

A desired product measure must satisfy

$$\mu\left(\prod_{i=1}^{n} A_i \right) = \prod_{i=1}^{n} \mu_i(A_i), \qquad 0 \cdot \infty := 0. \tag{2}$$

The plan is to define a premeasure on finite disjoint unions of rectangles, then apply Carathéodory's extension theorem.

Source page(s): 5–6

2 Construction of product measure

Theorem (reference Proposition 3.1.1). If each μ_i is sigma-finite, there exists a unique measure $\mu = \bigotimes_{i=1}^{n} \mu_i$ on $\mathcal{F}$ with the rectangle property. Let $\mathcal{A}$ be the finite disjoint unions of measurable rectangles. Define

$$\mu_0 \left(\bigsqcup_{j=1}^{m} \prod_{i=1}^{n} A_{j,i} \right) = \sum_{j=1}^{m} \prod_{i=1}^{n} \mu_i(A_{j,i}). \tag{3}$$

Supplement: Algebra and well-definedness homework

Intersecting rectangles gives a rectangle. A difference of rectangles splits into finitely many disjoint rectangles by selecting the first coordinate that lies outside the second rectangle. This yields complements and finite unions in $\mathcal{A}$. A common finite refinement of any two representations, together with finite additivity in each coordinate, shows that the displayed value is independent of the representation.

Source page(s): 7–10

3 Countable additivity by dimension reduction

It suffices to prove the identity when a rectangle is partitioned into countably many rectangles:

$$\prod_{j=1}^{n} A_j = \bigsqcup_{i=1}^{\infty} \prod_{j=1}^{n} A_{i,j}. \tag{4}$$

Induct on n, the case $n = 1$ being ordinary countable additivity. Fix $x \in \prod_{j<n} A_j$ and let

$$I_x = \left\{ i : x \in \prod_{j<n} A_{i,j} \right\}. \tag{5}$$

Taking the section at x gives $A_n = \bigsqcup_{i \in I_x} A_{i,n}$. Consequently

$$\mathbf{1}_{\prod_{j<n} A_j}(x) \mu_n(A_n) = \sum_{i=1}^{\infty} \mathbf{1}_{\prod_{j<n} A_{i,j}}(x) \mu_n(A_{i,n}). \tag{6}$$

If x lies outside the left rectangle's first $n-1$ factors, every active right rectangle has empty last factor; thus the identity holds there too, with the zero-times-infinity convention. Integrate with respect to the already constructed $\mu' = \bigotimes_{j<n} \mu_j$ and apply MCT to the nonnegative series. The rectangle property in dimension $n-1$ yields

$$\prod_{j=1}^{n} \mu_j(A_j) = \sum_{i=1}^{\infty} \prod_{j=1}^{n} \mu_j(A_{i,j}). \tag{7}$$

This proves the required premeasure property. Carathéodory gives existence. Choose increasing finite-measure exhaustions in each coordinate; their product rectangles exhaust Ω and have finite mass. The uniqueness theorem then applies to the pi-system of rectangles.

Supplement: The source's finite-measure reduction

The notes say to assume the measures finite without elaboration. One may localize to the finite-measure rectangles from the exhaustions and pass to an increasing limit. The section proof above also works directly with nonnegative extended quantities, provided products with zero are interpreted consistently.

Source page(s): 11–12

4 Fubini–Tonelli theorem

Let $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i = 1, 2$, be sigma-finite and let $\mu = \mu_1 \otimes \mu_2$. Let $f : \Omega_1 \times \Omega_2 \to \overline{\mathbb{R}}$ be product-measurable. If $f \geq 0$, the section integrals are measurable extended nonnegative functions and

$$\int f \, d\mu = \int_{\Omega_1} \left(\int_{\Omega_2} f(x, y) \, d\mu_2(y) \right) d\mu_1(x) \tag{8}$$

$$= \int_{\Omega_2} \left(\int_{\Omega_1} f(x, y) \, d\mu_1(x) \right) d\mu_2(y). \tag{9}$$

This is commonly called *Tonelli's theorem*. If $\int |f| \, d\mu < \infty$, the same equality holds with finite integrals; the sections are integrable for almost every outer coordinate. This is *Fubini's theorem*. On the exceptional outer-coordinate sets where a signed section integral is undefined, set that integral to zero. Moreover, either finite iterated integral of $|f|$ proves that f is integrable, by the nonnegative case (reference Theorem 3.2.2).

Original TODO: What is meant by product-measurable?

It means $f^{-1}(C) \in \mathcal{F}_1 \otimes \mathcal{F}_2$ for every Borel $C \subseteq \overline{\mathbb{R}}$. Measurability of each separate section alone is a different hypothesis and does not in general imply this. The reference combines the nonnegative and integrable cases under the name Fubini's theorem; the source's attribution of the names is retained here with the common distinction explained.

Source page(s): 13

Original TODO: Check section measurability

For fixed x, the sets E whose sections $E_x = \{y : (x, y) \in E\}$ lie in $\mathcal{F}_2$ form a sigma-algebra containing the rectangles. Hence every product-measurable set has measurable sections. Applying this to inverse images of Borel sets shows $y \mapsto f(x, y)$ is measurable (reference Lemma 3.2.1). The other coordinate is symmetric.

Source page(s): 14–16

5 Proof: rectangles, simple functions, and limits

For $f = \mathbf{1}_{A_1 \times A_2}$, direct computation gives

$$\int_{\Omega_1} \int_{\Omega_2} \mathbf{1}_{A_1 \times A_2}(x, y) \, d\mu_2(y) \, d\mu_1(x) = \mu_1(A_1)\mu_2(A_2) = \mu(A_1 \times A_2). \tag{10}$$

After extending the indicator identity to all measurable sets (next section), finite nonnegative linear combinations prove it for simple functions. If $s_n \uparrow f \geq 0$, MCT first in the inner integral, then in the outer integral, and also on the product space proves the identity for f. The section-integral functions are measurable as increasing limits of the corresponding simple-function integrals.

Source page(s): 17–20

6 Proof: general indicators via the pi–lambda theorem

For finite μ_1, μ_2, let $\mathcal{D}$ consist of product-measurable sets E for which $x \mapsto \mu_2(E_x)$ is measurable and

$$\int_{\Omega_1} \mu_2(E_x)\, d\mu_1(x) = \mu(E). \tag{11}$$

Rectangles belong to $\mathcal{D}$. If $E_i \in \mathcal{D}$ are disjoint, their sections are disjoint, so

$$\mu_2\left(\left(\bigcup_i E_i\right)_x\right) = \sum_i \mu_2((E_i)_x), \tag{12}$$

$$\int \sum_i \mu_2((E_i)_x)\, d\mu_1(x) = \sum_i \mu(E_i) = \mu\left(\bigcup_i E_i\right). \tag{13}$$

Measurability follows from nonnegative measurable summation and the integral equality from MCT. For complements,

$$\mu_2((E^c)_x) = \mu_2(\Omega_2) - \mu_2(E_x). \tag{14}$$

All terms are finite, and integration gives $\mu(\Omega) - \mu(E) = \mu(E^c)$. Thus $\mathcal{D}$ is a lambda-system containing the generating pi-system of rectangles. Dynkin's theorem proves the indicator identity.

Original TODO: Review the pi–lambda proof technique

The defining class must include both the measurability of the section-mass map and its integral identity. Establish the result on the easy generating pi-system, prove the desired class is a lambda-system, then invoke Dynkin. Merely declaring the class to be a sigma-algebra would skip the substantive step.

Supplement: Passing from finite to sigma-finite measures

Choose $X_n \uparrow \Omega_1$ and $Y_n \uparrow \Omega_2$ with finite coordinate measures. The restricted measures $\mu_{1,n}(A) = \mu_1(A \cap X_n)$ and $\mu_{2,n}(B) = \mu_2(B \cap Y_n)$ have product $E \mapsto \mu(E \cap (X_n \times Y_n))$, by equality on rectangles and uniqueness. Apply the finite result and then MCT to

$$x \longmapsto \mathbf{1}_{X_n}(x)\mu_2(E_x \cap Y_n) \uparrow \mu_2(E_x). \tag{15}$$

This supplies the localization step omitted in the handwritten proof (reference proof of Theorem 3.2.2).

Original TODO: Review the integrable case

Apply the nonnegative theorem to $|f|$ to see that $\int |f(x,y)|\, d\mu_2(y) < \infty$ for almost every x. Apply it to f^+ and f^- and subtract their finite total integrals. Their section integrals are

finite outside a common null set; hence subtraction there is legitimate. Repeat in the other order. This justifies the source's abbreviated signed-integral step without requiring every section to be integrable.

Source page(s): 21–22

7 Midterm preparation recorded in the source

The handwritten announcement describes four questions: a first true/false question and three further questions with one to four subparts. Coverage is Chapters 1 through 3.2. Listed preparation topics are:

1. Decide whether a set function is a measure and whether it is sigma-finite.
2. Determine the sigma-algebra generated by a specified class.
3. Check measurability and compute Lebesgue integrals on simple spaces.
4. Decide whether a limit can be exchanged with an integral using MCT, Fatou, or DCT.
5. Decide whether an iterated integral equals the product-space integral for a concrete function.
6. Distinguish Riemann integration from improper integration.

The source describes the exam as easier than the homework. This is a transcription of the note's announcement, not a verification of a current exam policy.